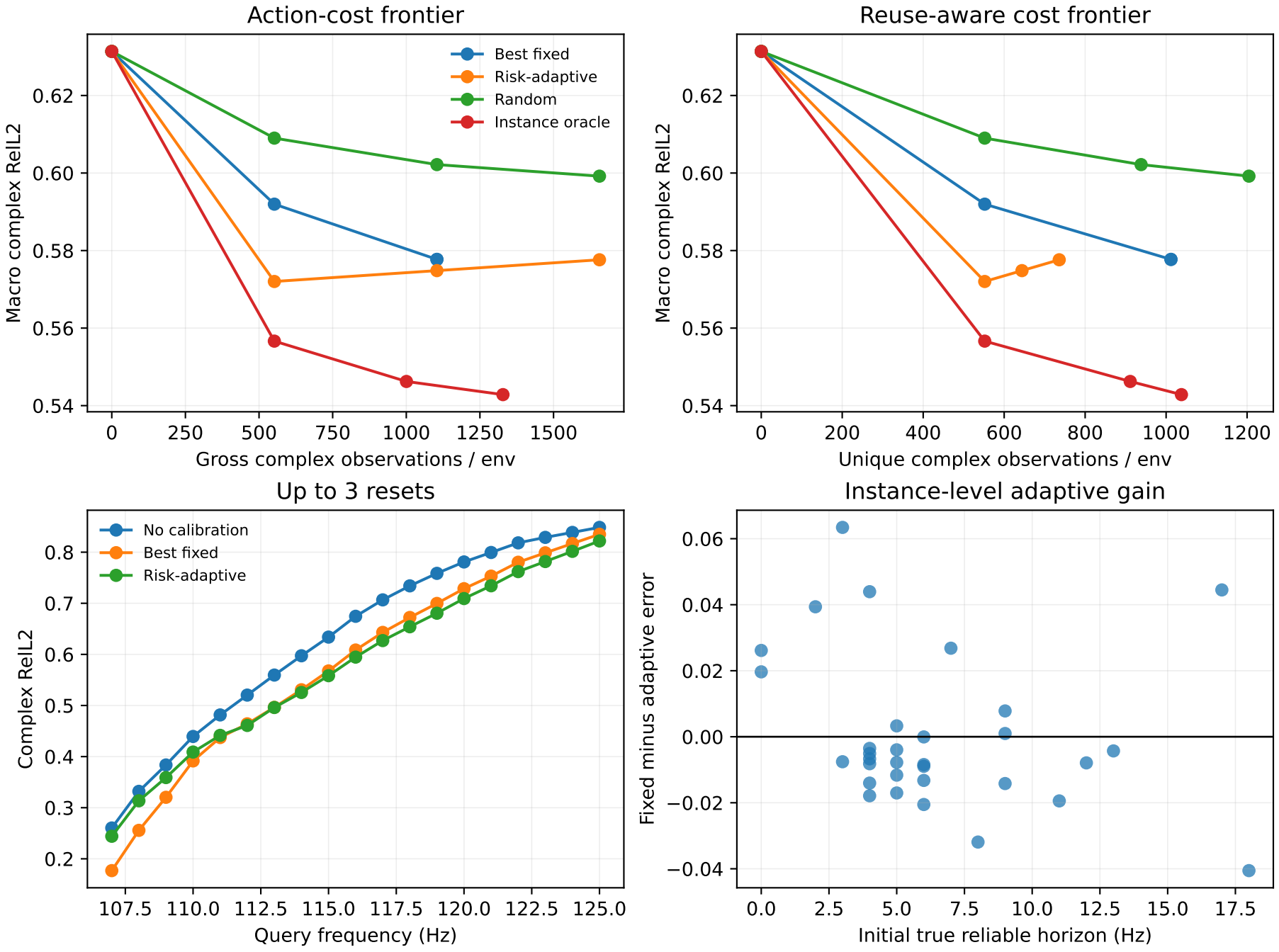
Action-cost frontier
Best fixed
Risk-adaptive
Random
Instance oracle
Macro complex RelL2
0.62
0.60
0.58
0.56
0.54
0
250
500
750
1000
1250
1500
Gross complex observations / env
Reuse-aware cost frontier
Macro complex RelL2
0.62
0.60
0.58
0.56
0.54
0
200
400
600
800
1000
1200
Unique complex observations / env
Up to 3 resets
No calibration
Best fixed
Risk-adaptive
Complex RelL2
0.8
0.7
0.6
0.5
0.4
0.3
0.2
107.5
110.0
112.5
115.0
117.5
120.0
122.5
125.0
Query frequency (Hz)
Instance-level adaptive gain
Fixed minus adaptive error
0.06
0.04
0.02
0.00
−0.02
−0.04
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Initial true reliable horizon (Hz)